

# Generation & Validation of a Mechanical Bidirectional Sweep Signal to Bolster Human T-Cell Proliferation for Cancer Immunotherapy

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## Introduction

### Motivation:

- During exercise, the body experiences a wide range of vibrations that influence different physiological processes.
- Low-Intensity Vibration (LIV) delivers mechanical oscillations in the range of ~10-150Hz to biological systems.
  - Key parameters include: displacement, acceleration, duration, refractory period, and number of treatments.
- Previously studied effects of 0.7g, 30 Hz vibrations (2×1hr/day for 5 days) on T-cell proliferation to enhance biomanufacturing yields for CAR-T therapies.

### Objective:

- To develop a Frequency and Amplitude Modulated Sweep (FAMS) signal that applies a band of several frequencies, providing a more physiologically relevant model of the vibrations experienced during exercise.

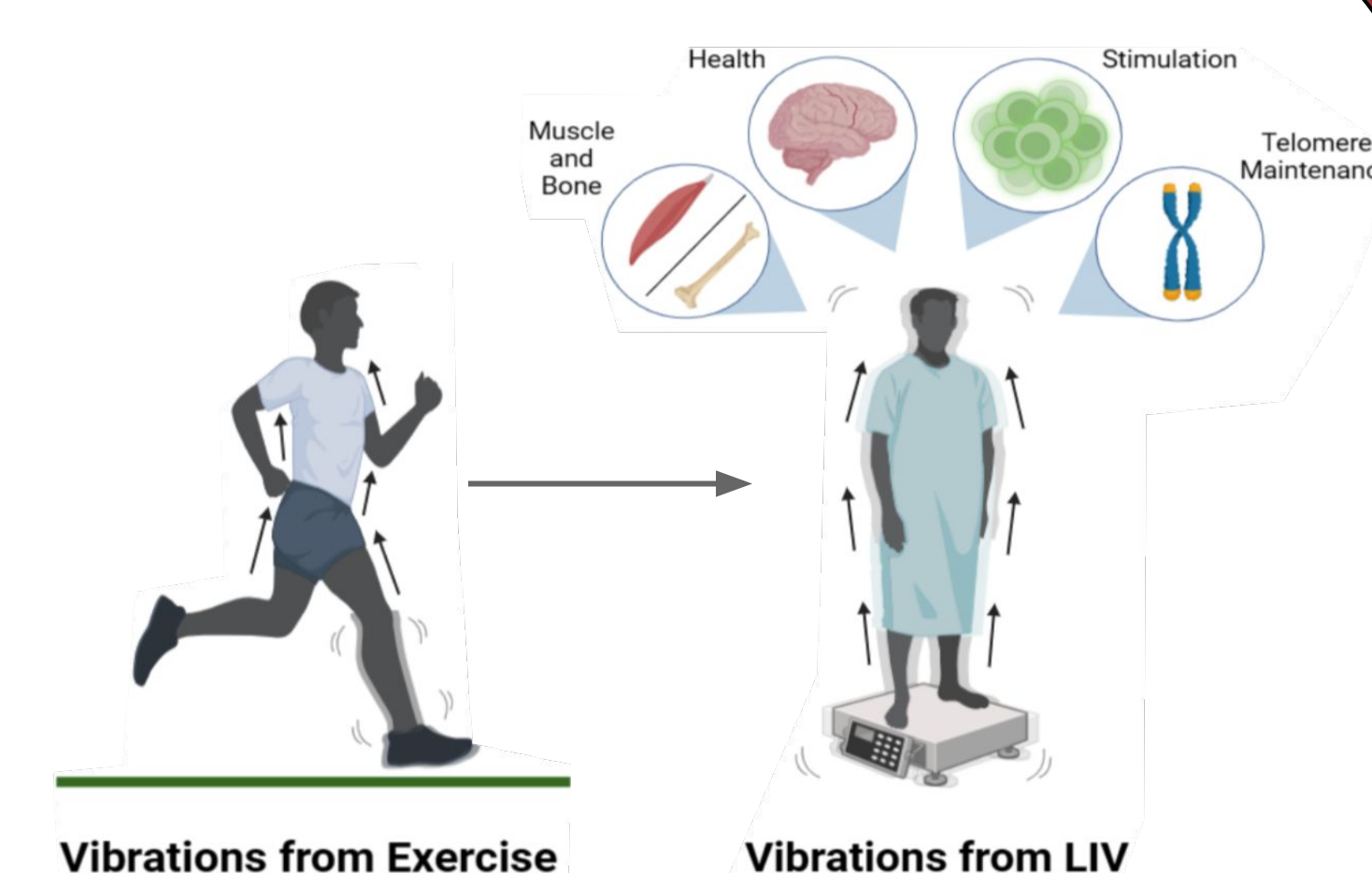


Fig 1. Various LIV applications.

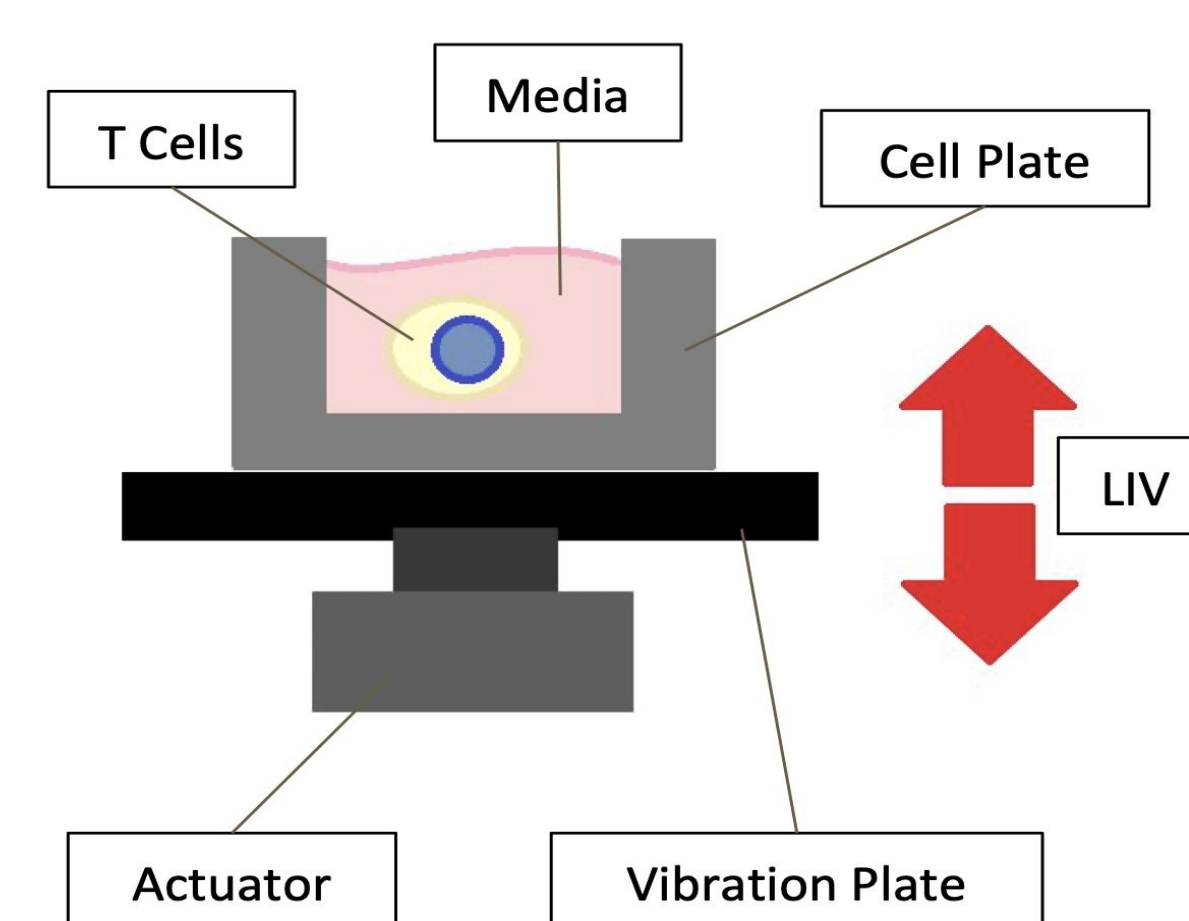


Fig 2. LIV experimental setup for stimulating suspension cell.

## Data Analysis

### Signal Validation

#### Features:

- Acceleration measured using *enDAQ Slamstick*, piezoelectric accelerometer.
- Python script features several libraries: *enDAQ*, *plotly*, *pandas*, *numpy*, *scipy*.
- Python Script Quality Control:
  - Acceleration & Displacement Visualization
  - Peak Acceleration Detection
  - Individual Chirp Analysis
  - Ramp Time, Rate of Change, and Frequency Verification
  - Noise detection

#### Results:

- Visualized quality control figures and reports using *plotly*.

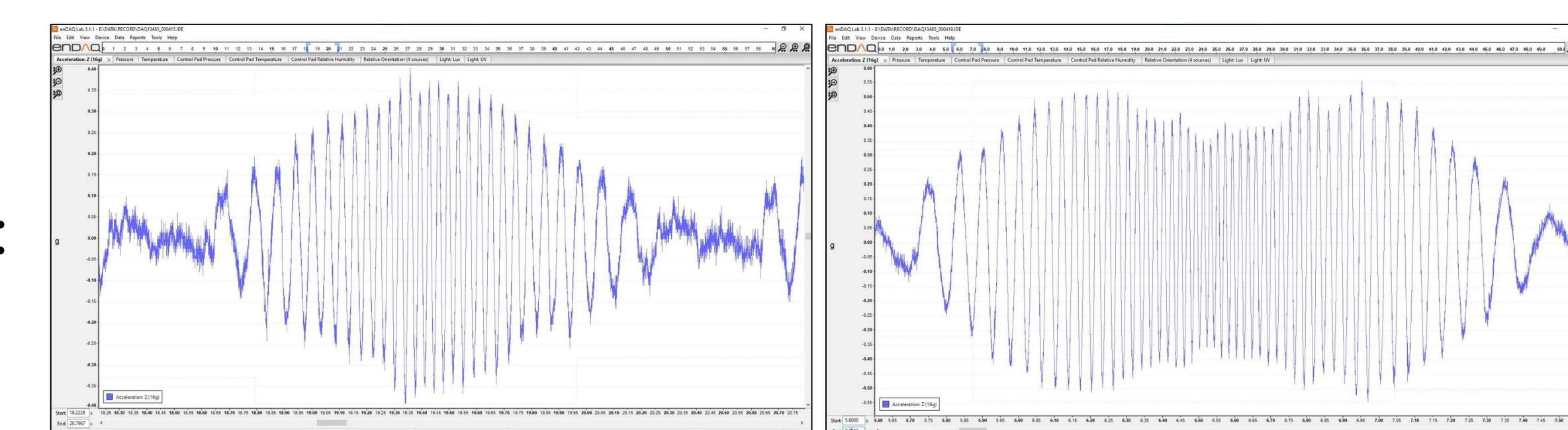


Fig 4. *enDAQ* software showing a successful sweep from 0–30 Hz (left) and an unexpected dip in acceleration at higher frequencies, peaking at 50 Hz (right).

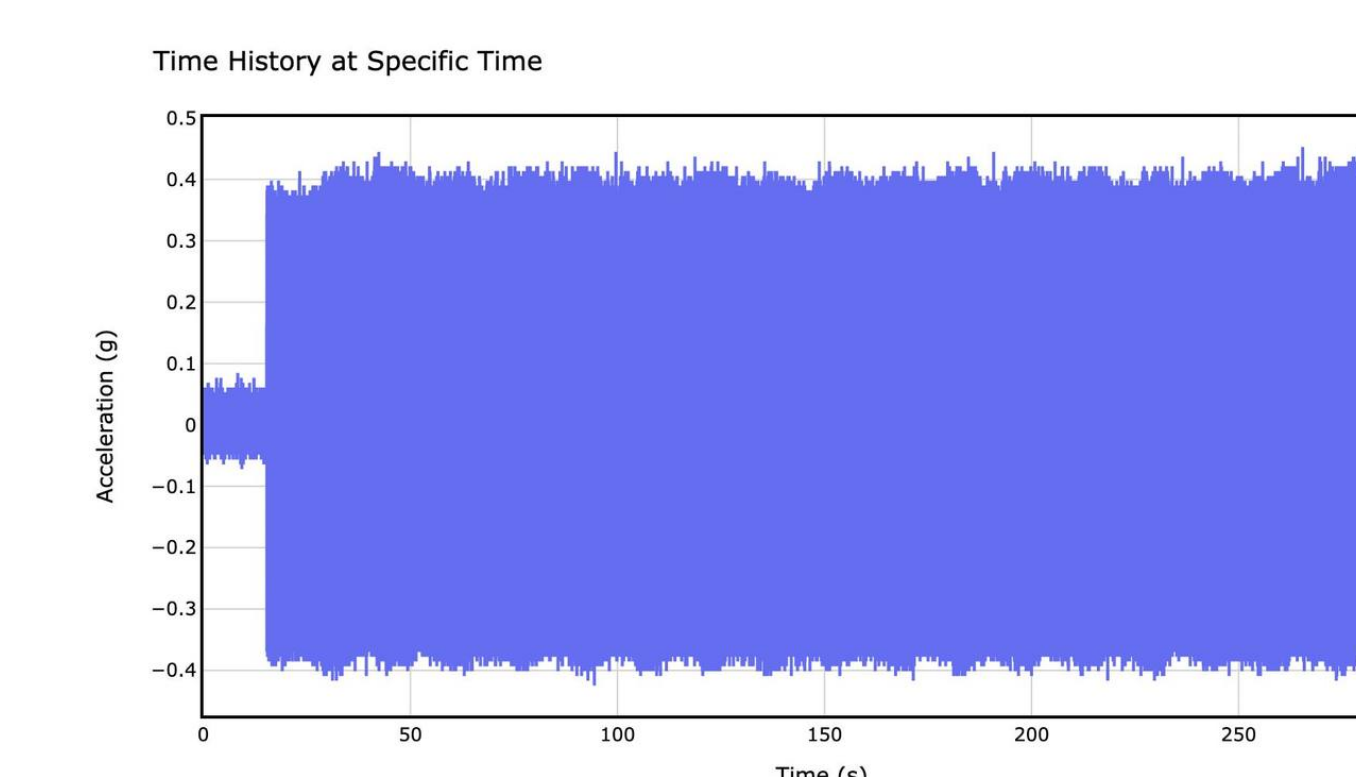


Fig 5. Signal validation using a Python script to display a test, static 30 Hz signal at 0.7g p-p.

## Methods & Results

### LIV Device & FAMS Signal Generation

#### System Features:

- LabVIEW software
- Microcontroller—Arduino UNO
- Cylindrical Linear Voice Coil Actuator (solenoid mechanism)
- Triple-Axis Accelerometer
- Rubber Bands (to secure culture flasks and plates)

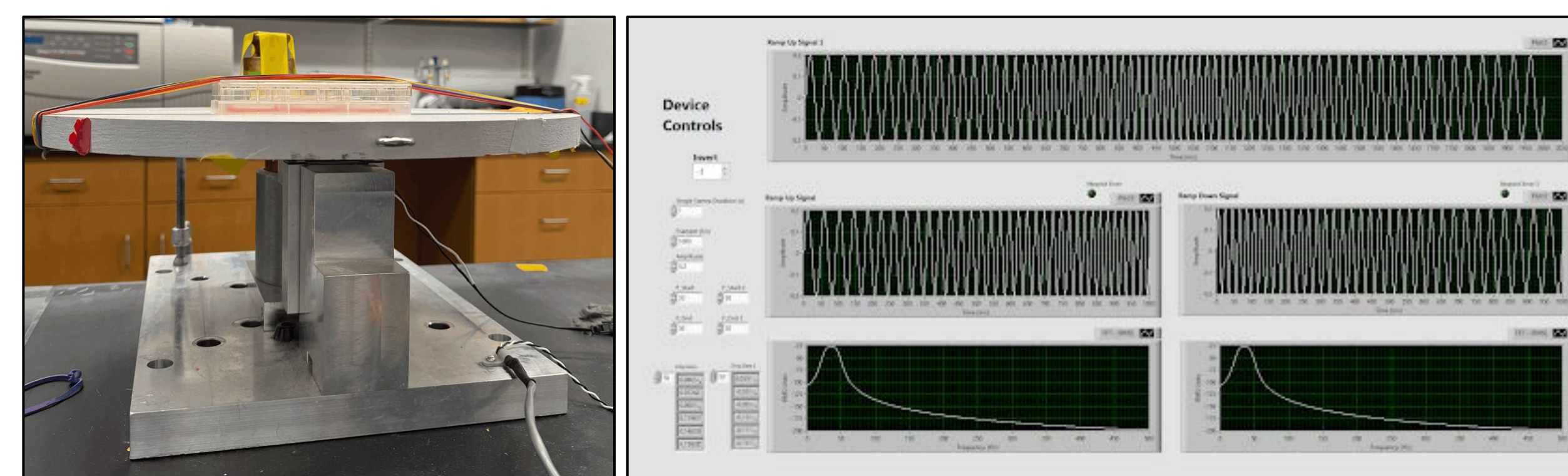


Fig 3. Tabletop device setup (left) and LabVIEW virtual interface (right).

#### Results:

- A bilateral sweep signal is generated via a LabVIEW VI, then converted, amplified, and sent to the actuator to produce vibrations at the desired frequencies and accelerations.
- The Labview VI allows custom sweep signal inputs for selecting frequencies, amplitude, timing, and sampling rate.
- The device can successfully produce the expected vibration profile from 0 to 35 Hz.
- However, oscillating artifacts are seen at frequencies of 36 Hz and higher.

## Conclusion & Future Research

- The current LIV cell stimulation device successfully performs a bilateral sweep from 0 to 35 Hz; however, further development is needed to achieve consistent vibration delivery in the 36–50 Hz range.
- The Python-based quality control script requires continued development to ensure consistent delivery of sweep frequencies and accelerations, and to verify that vibration intensity does not exceed 0.9g (peak-to-peak).
- Integrating live acceleration feedback with a calibrated accelerometer will streamline testing by removing the need to export *Slamstick* data to a computer.
- Upcoming in vitro study:** T-cells will be stimulated with the FAMS signal (20–50 Hz) in two 1-hour sessions per day over 5 days, mirroring the schedule of foundational studies.

## References

- <https://bpb-us-e1.wpmucdn.com/you.stonybrook.edu/dist/c/4606/files/2020/11/BM-ES-2020-Presentation.pdf>
- Low intensity mechanical signals promote proliferation in a cell-specific manner: Tailoring a non-drug strategy to enhance biomanufacturing yields. M. Ete Chan, Lia Strait, Christopher Ashdown, Sishir Pasumathy, Abdullah Hassan, Steven Crimarco, Chanpreet Singh, Vihitaben S Patel, Gabriel Pagnotti, Omor Khan, Gunes Uzer, Clinton T Rubin *bioRxiv* 2023.07.05.547864; doi: <https://doi.org/10.1101/2023.07.05.547864>

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